Weather Frame — Design Documentation

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Overview

Change of electronics, 13 Oct 2026: JLCPCB can only place the ESP32-S3 module with Standard assembly, and the build has to work without any soldering. The electronics are therefore now a ready-made Seeed XIAO ePaper Display Board EE05 (ESP32-S3, 24-pin panel connector, battery charger), which replaces both the custom PCB and the Waveshare driver HAT. The firmware runs on it as `weatherframe_ee05.yaml`. The custom-PCB sections below remain as the design record for a possible later revision; the frame is being adapted to the EE05.

The Weather Frame is a battery-powered weather station built into a wall picture frame: a Waveshare 7.5″ e-paper panel (800 × 480, 4-level grayscale) shows the current weather, an hourly chart and a 5-day forecast for Budapest, refreshed every 15 minutes from Open-Meteo over Wi-Fi. It runs for about two months on one charge of a 3000 mAh LiPo and charges over USB-C.

Requirements the design was built against:

- Looks like a picture frame on the wall: no visible cables, no visible electronics, hangs on two screws; a fold-out stand lets it stand on a cupboard instead.
- Rechargeable: 1S LiPo inside, charged over USB-C, no power supply on the wall.
- Wi-Fi only; no hub or server needed. Weather data straight from the Open-Meteo API.
- Indoor temperature and humidity on the display as well (SHT40 on the board).
- Manufacturable from one vendor: PCB and assembly at JLCPCB, frame parts at JLC3DP, no hand-placed SMD parts.
- German labels, metric units, quiet hours at night; all of this is configuration, not code.

The project has three parts, each generated from one source file so they stay consistent:

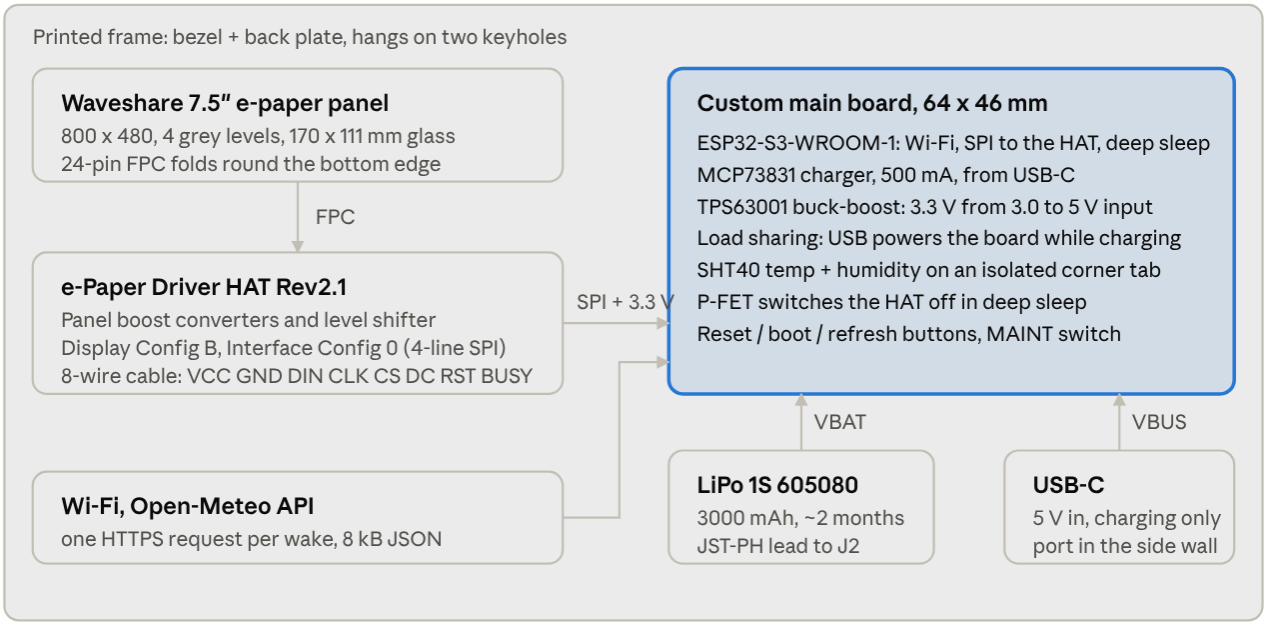
Part	Source of truth	Output
Main PCB (64 × 46 mm, ESP32-S3)	`hardware/kicad/weatherframe/gen/design.py`	KiCad schematic + board, JLCPCB Gerbers / BOM / CPL
Firmware	`firmware/weatherframe.yaml` + `weatherframe/weatherframe.h`	ESPHome build for the ESP32-S3

Part	Source of truth	Output
Frame (bezel + back plate)	`hardware/mechanical/frame_geom.py`	Fusion 360 model, STEP / STL for printing

System architecture

The custom board is the only active electronics besides the Waveshare driver HAT: it owns power, charging, Wi-Fi and the sensor, and talks to the HAT over a 4-wire SPI cable. The HAT stays as bought, because it carries the panel's high-voltage boost circuits and the 24-pin ZIF connector that would be hard to replicate and test on a first board.

The custom board charges the battery, powers everything and drives the panel through the HAT



system architecture · 5 parts inside the frame

The panel's ribbon folds round the bottom edge of the glass into the HAT, which sits behind the glass on four standoffs; the custom board sits behind the right side of the glass with its USB-C port in the frame's side wall, and the battery lies flat behind the upper left.

Electronics design decisions

Every part is in JLCPCB's assembly catalogue, with "basic" parts wherever a basic part would do, so the board can be assembled without hand soldering. Only J3 (the HAT cable) and the four mounting holes are left unpopulated.

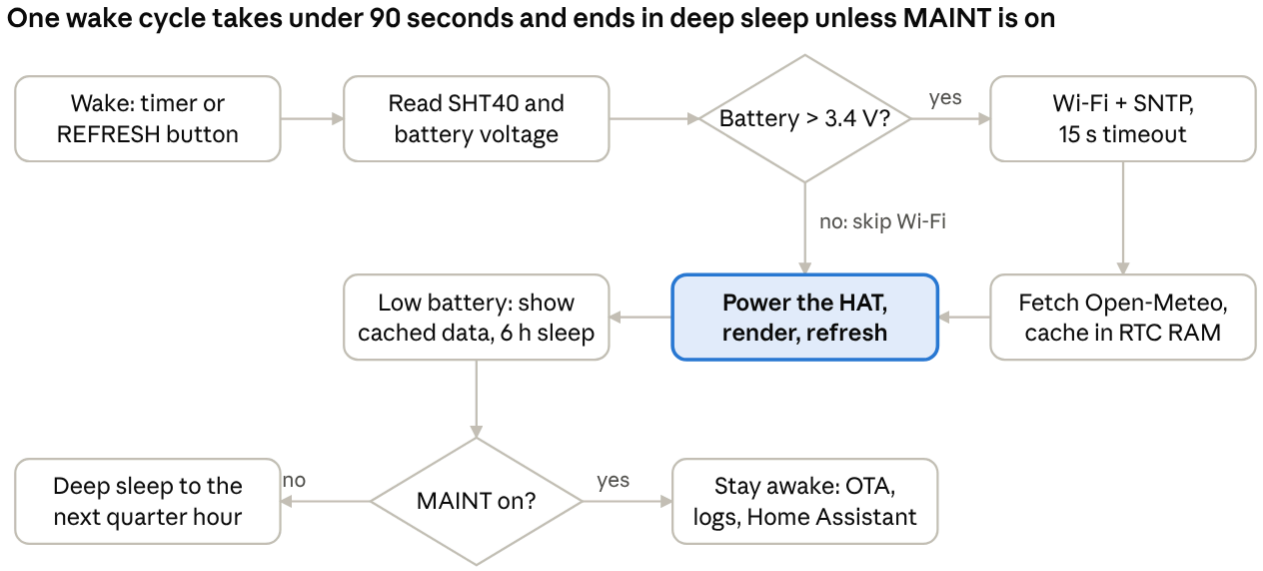
Block	Decision	Why
MCU	ESP32-S3-WROOM-1-N16R8 module, ESP-IDF framework	Native USB for flashing without a UART bridge, 8 MB PSRAM for the 800 × 480 frame buffer and fonts, module with a certified antenna; the antenna end overhangs the board edge as the datasheet asks.
3.3 V rail	TPS63001 buck-boost, 1.5 µH, 2 × 22 µF out	A LiPo swings from 4.2 V to 3.0 V and USB is 5 V; a buck-boost holds 3.3 V over the whole range, so the panel and the radio behave the same at a full and at an empty battery. PS/SYNC is tied low for power-save mode, which keeps the quiescent current low during deep sleep.
Charger	MCP73831-2, PROG = 2 kΩ → 500 mA	Simple, single-cell, cheap; 500 mA charges a 3000 mAh cell in about 7 hours and is well below the 0.5 C limit of a 605080 pouch. A red LED on STAT shows charging.
Load sharing	B5819W Schottky from VBUS plus AO3401A P-FET from VBAT	When USB is present the diode feeds the regulator and the FET's gate is pulled high, so the battery is not loaded while it charges; without USB the FET conducts and bypasses its own body diode. Standard "ideal diode" arrangement with two cheap parts.
USB-C	HRO TYPE-C-31-M-12 (16-pin), 5.1 kΩ on CC1/CC2, USBLC6-2SC6 ESD	The 5.1 kΩ pull-downs make a USB-C charger supply 5 V; D+/D− go to the ESP32-S3's native USB for flashing. The USBLC6 protects the data lines against ESD at the port.
Battery sensing	1 MΩ / 1 MΩ divider with 100 nF into GPIO5 (ADC)	The divider halves 4.2 V to 2.1 V, inside the 12 dB attenuation range; 1 MΩ keeps the standing drain at 2 µA, and the capacitor gives the ADC a low-impedance source.

Block	Decision	Why
USB and charge status	100 kΩ / 150 kΩ dividers into GPIO4 and GPIO15	The ESP32-S3 is not 5 V tolerant; 5 V → 3.0 V stays above the 2.475 V logic-high threshold and below the 3.6 V absolute maximum across the 4.75–5.25 V USB range.
E-paper power	AO3401A P-FET on the HAT's 3.3 V, 100 kΩ pull-up, driven by GPIO6	The HAT draws current even when idle; switching it off in deep sleep is the single largest contribution to the two-month runtime. The pull-up keeps it off while the ESP32 is in reset or asleep.
Indoor sensor	SHT40-AD1B on an isolated corner tab with a thermal slot, 4.7 kΩ I²C pull-ups	The ESP32 and the regulator warm the board by a degree or two; the slotted tab and no copper pour around the sensor keep it near room temperature. The firmware reads it before Wi-Fi starts for the same reason.
Controls	RESET and BOOT tactile switches, REFRESH button on GPIO13 (RTC wake), MAINT slide switch on GPIO14	REFRESH wakes the device from deep sleep for an immediate update; MAINT keeps it awake so OTA updates and logs work without timing a 90-second window.
HAT connector	J3, 1 × 9 header footprint, hand-soldered	Kept as plain 2.54 mm holes so any cable can be used; in the frame the HAT's 8-wire cable is soldered in directly.
Board	64 × 46 mm, 2 layers, 0.15 mm clearance, 0.2 mm tracks, 0.3 mm vias, M2.5 holes	Fits JLCPCB's standard 2-layer process with no extra-cost options; small enough to sit beside the HAT behind the glass.

Firmware design

The firmware is an ESPHome configuration (ESP-IDF framework, validated with ESPHome 2026.9.1) plus one C++ header for the Open-Meteo parsing and the screen layout. ESPHome was chosen over hand-written ESP-IDF code because it gives OTA updates, Wi-

Fi handling, the e-paper driver, fonts and a Home Assistant API for free, and because the whole behaviour is readable in 220 lines of YAML.



wake cycle · every 15 minutes, aligned to :00 / :15 / :30 / :45

Design choices that drive the battery life and the user experience:

- **Everything is time-boxed.** Wi-Fi gets 15 s, the clock 3 s, the HTTP request 10 s, the panel refresh 30 s, and `run_duration: 90s` is a hard ceiling. A failed step shows the cached data instead of waiting.
- **The weather is cached in RTC memory** across deep sleep, so a Wi-Fi outage keeps the last forecast on the screen with a "wifi-off" icon rather than a blank display.
- **Quiet hours.** `wf::next_sleep_ms()` aligns wakes to quarter hours and skips 00:00–06:00 (configurable), which saves about a quarter of the daily energy.
- **Low-battery mode** below 3.40 V without USB: no Wi-Fi, a battery-alert icon, and a 6-hour sleep so the cell is never deep-discharged.
- **Sensors first.** The SHT40 is read before Wi-Fi starts so the reading is not biased by the board warming up.
- **The panel is powered only while rendering** through the P-FET (`enable_pin` of the display component).
- **Rendering** uses Inter (Google Fonts) at four sizes with 1-bit glyph tables limited to the characters actually used, plus Material Design Icons for the weather symbols; the layout is shared with `preview.yaml`, which renders the same screen in an SDL window on the Mac for layout work without hardware.
- Location, language, units, interval and quiet hours are substitutions at the top of `weatherframe.yaml`; the pin map is listed there too and must match `design.py`.

Mechanical design

The frame is three printed parts, 192.8 × 131.5 × 24 mm overall (27 mm including the folded stand), that screw together with four M3 screws into heat-set inserts: a **bezel** that holds the glass and a **back plate** that carries all the electronics, and a fold-out stand for standing it on a cupboard. The glass is clamped between the bezel's 2.5 mm lip and foam-backed press pads on the back plate, so no adhesive touches the panel and it can be replaced.

`frame_geom.py` is the single source: it expresses the whole design as a list of boxes, cylinders and slots in millimetres. The same list is built by CadQuery (STEP/STL, interference checks, previews) and by the Fusion 360 script, which was executed through the Fusion MCP connector; both produce identical volumes (bezel 59.4 cm³, back plate 86.2 cm³). Change a number in the `P` dict and re-run either.

Element	Value	Reasoning
Window	active area + 0.8 mm per side, 1 mm 45° chamfer	Shows the full image with a thin white panel border; the chamfer hides the print layer lines at the edge.
Panel recess	170.2 × 111.2 mm + 0.4 mm clearance, 1.5 mm deep in a 4 mm face	Locates the glass in-plane; the active area is centred in the window although the panel's borders are 3.5 mm on three sides and 9.8 mm on the ribbon side.
Rim	14 mm sides and top, 18 mm bottom	The bottom rim hides the ribbon fold and the HAT; 14 mm reads as a picture frame rather than a device. Outer corners R6.
Cavity	17 mm deep, 2.5 mm walls	Set by the HAT: 7.5 mm standoffs, 1.6 mm board, 6 mm JST connector on top, 2.2 mm air to the glass. The 40-pin header on its back sinks 1.5 mm into the plate.
HAT mount	4 × M2.5 standoffs, 58 × 23 mm (Pi-Zero pattern), ribbon edge 11.6 mm from the glass edge	Measured from the Rev2.1 board; the 24 mm ribbon has ~2 mm to spare after the fold and 3 mm insertion.

Element	Value	Reasoning
PCB mount	4 × M2.5 standoffs 5 mm, components toward the plate	Puts the USB-C port in the right wall (wall thinned to 1 mm there so the plug seats), the MAINT lever under a slot and the three buttons under Ø3 holes in the plate.
Battery	605080 pocket, 1 mm clearance, 3 mm fence with a lead gap	Lies flat behind the upper left; the fence stops it sliding onto the HAT or the pads.
Panel pads	6 pads, 3 mm wide, topped with 1 mm foam	Placed on the panel borders where no electronics sit: top both sides of the keyholes, bottom both sides of the ribbon, lower left, upper right.
Hanging	2 keyholes at x = ±45 mm, slot upward, 2 mm bosses inside	Two screws keep the frame level; the slot runs upward because the frame slides down onto the screw heads.
Vents	5 slots over the SHT40 corner	Lets room air reach the sensor through the plate. Flip-out stand: a separate 60 × 75 × 3 mm leg hinged 10 mm above centre on the back over a 36.8 mm wide three-knuckle hinge (M3 × 40 pin, nyloc nut for friction) with a built-in stop at 45° opening, so the frame stands leaning back 25° on its rear-bottom edge with a 59 mm footprint; folded it stands 3 mm proud. Swept 0–50° in CadQuery: free to 45°, contact from 46°. Colour: white, to sit quietly on a white woodchip wall. Labels: RESET, BOOT and MAINT are engraved left of their openings and REFRESH right of its hole (as seen from behind), 3 mm bold, 0.5 mm deep.
Printing	all walls ⩾ 1 mm, pockets ⩾ 1.5 mm plate left, no overhangs on the plate	Suitable for MJF nylon or SLA at JLC3DP, or an FDM printer with the bezel face down.

Problems encountered and how they were resolved

All of these were found by review, simulation or geometry checks before anything was ordered; none required a board re-route.

Area	Problem	Resolution
Electronics	The USB-present and charger-status dividers were 100 kΩ / 100 kΩ: 5 V → 2.5 V, which sits on the ESP32-S3's logic-high threshold (0.75 × 3.3 V = 2.475 V) and reads as undefined on a 4.75 V supply.	Bottom resistors R6 and R11 changed to 150 kΩ (LCSC C25755): 5 V → 3.0 V, 2.85–3.15 V over the USB range. Same footprint, so only the values, the BOM and the schematic changed.
Electronics	The TPS63001 pin mapping in `design.py` had to be trusted blindly: FB tied to VOUT and VINA to VIN are easy to get wrong on this part.	Verified against the TI datasheet pin table and the KiCad library symbol extracted from the schematic: all ten pins are correct, PS/SYNC low = power-save.
Electronics	J3 was a vertical 2.54 mm header; with the board components facing the back plate, a header plus a cable housing needs ~14 mm, but only 5 mm exist.	The HAT's 8-wire cable is soldered directly into J3 for this revision; a right-angle JST-PH is noted for r2.
PCB	DRC reports 4 silkscreen-clipped warnings and 1 footprint mismatch.	Both are expected: the ESP32 antenna and the USB-C connector sit at the board edge by design, and the SHT40 footprint is a local copy without the central pad. 0 errors, 0 unconnected.
Firmware	The sandbox's ESPHome (2025.7) does not know the `epaper_spi` platform; a current one needs Python ≥ 3.11.	Installed ESPHome 2026.9.1 in a Python 3.12 venv with uv; the config validates. Google Fonts and the ESP-IDF download are blocked by the proxy, so fonts came from the cache of the earlier run and the compile is left to the Mac.

Area	Problem	Resolution
Tooling	The previous Claude Code session stopped when its API credit ran out, leaving no notes on where it stood; KiCad is only installed on the Mac and the Fusion MCP was not reachable from the shell.	State was reconstructed from file timestamps and reports; a `build.sh` was added so the KiCad pipeline is one command; the Fusion MCP connector was added to the desktop app, which made Fusion scriptable from the session.
Mechanical	First layout had the battery under a panel press pad, the PCB over a corner screw boss, and the keyholes inside the battery pocket.	Found by CadQuery interference checks between the printed parts and reference volumes; the battery moved above the HAT, the PCB up by 8 mm with the MAINT switch reached through a slot instead of the wall, keyholes moved to x = ±45 mm, y = 54 mm.
Mechanical	The panel recess was cut 1 mm into the cavity instead of into the face, leaving a 2 mm face.	Face thickness set to 4 mm with the 1.5 mm recess cut into it (2.5 mm lip).
Mechanical	The HAT's JST connector (6 mm tall) would have been 0.7 mm from the glass with the HAT on a fence.	HAT geometry re-measured from photos: four M2.5 holes, header on the far side from the ribbon. Four standoffs, the header pocketed 1.5 mm into the plate, cavity +1 mm → 2.2 mm clearance.
Mechanical	The ribbon length (24 mm) is tight for folding round the glass into the ZIF.	HAT ribbon edge placed 11.6 mm below the glass edge: fold + z-step + 3 mm insertion ≈ 21.6 mm. One parameter (`hat_center`) if it needs adjusting after the first fit.

Area	Problem	Resolution
Mechanical	The Waveshare spec PDF could not be downloaded through the proxy, and Chrome's PDF viewer could not be driven.	Read the mechanical drawing through the Google Docs viewer in the built-in browser; the border and ribbon dimensions in `frame_geom.py` come from that page. Ordering: JLCPCB restricts the ESP32-S3 module to Standard assembly ($25 setup, break-off rails that clash with the antenna overhang), and the build must need no soldering. Resolution: switched to a Seeed EE05 board, which plugs straight into the panel ribbon; firmware ported, frame holder to follow.

Open items and ideas for revision 2

Nothing below blocks ordering revision 1; the first three are checks to do when the parts arrive.

- [ ] Compile and flash the firmware on the real board; confirm the 15-minute cycle completes in under 90 s and measure the deep-sleep current (target under 100 µA with the HAT switched off).
- [ ] Fit check of the back plate: HAT on its standoffs, ribbon reach into the ZIF, battery in its pocket, USB-C plug seating through the side wall.
- [ ] Confirm the keyhole spacing (90 mm) against the wall screws and that the frame hangs plumb with the battery on the left.
- [ ] Battery lead: a 605080 cell usually comes with a 50–60 mm lead; J2 is about 80 mm away, so order the cell with a longer lead or add an extension.

Ideas for r2:

- Replace J3 with a right-angle JST-PH 8-pin so the HAT's cable plugs in; or move the HAT's boost circuit onto the main board and drive the panel's 24-pin ZIF directly, which would remove the HAT, 8 mm of thickness and the cable.
- A 2-pin JST-PH for the battery is standard but not polarised the same way by every supplier; a note on the silkscreen ("+ left") would prevent a reversed cell.
- Add a photoresistor or the SHT40's data to Home Assistant through the API while in MAINT mode.

- Consider 10 mm rims with a wood-look print finish once the first frame confirms the proportions.